

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)



PRESENTS



Ai - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaia School of Engineering
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PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME (AS SUBMITTED ON PORTAL)		Demo Buddies	
PROBLEM STATEMENT TITLE		Tracking of Funds within Bank for Fraud Detection	
	NAME	GENDER	AREA OF EXPERTISE
Member 1	Somya Upadhyay	Female	Machine Learning & Data Science
Member 2	Sagar Agarwal	Male	Full-Stack Development & AI/ML
Member 3	Tathagata Sen	Male	Graph Analytics & Database Engineering
Member 4	Aviral Sharma	Male	Backend Systems & Cloud Deployment

IDEA TITLE

GraphSentinel

AI-Powered Federated Fund Flow Tracking & Fraud Detection System

DETECT

Graph ML identifies layering, round-trips, structuring, dormant account abuse & KYC mismatches - all five PS3 threat patterns - in real time as transactions stream in

EXPLAIN

Every alert ships with a SHAP-powered causal narrative: a plain-English fund trail showing exactly which transactions triggered the flag and why

REPORT

Auto-generates FIU-IND compliant STR/CTR evidence packages in goAML XML format - reducing investigator effort from 4 hours per report to under 10 minutes

KEY INNOVATION: Federated GNN - fraud intelligence grows smarter across all PSBs without sharing raw customer data. Only privacy-preserving model gradients are exchanged.

Proposed Solution - How it addresses the problem:

- Fraud that spans multiple accounts, branches, or banks is invisible to rule-based systems - our model builds a real-time transaction graph and applies temporal GNNs to see the complete multi-hop picture.
- Unlike existing AML tools that output a risk score alone, our prototype produces a full fund-flow narrative and an auto-generated FIU report, closing the gap between detection and prosecution.

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

Layer 1 Real-Time Graph Construction	Apache Kafka ingests NEFT/RTGS/UPI/Core Banking streams. Each transaction is a directed edge; each account is a node with KYC attributes. The graph is updated in <2500 ms per transaction batch.
Layer 2 Temporal Pattern Detection	EvolveGCN + DFS-based cycle detection on sliding 6-hour windows. Flags: layering chains (3+ hops in <4hrs), round-trips (funds return to origin account), structuring (<₹10L repeated clusters), velocity spikes.
Layer 3 Behavioural Profile Deviation	LSTM autoencoder models each customer's 12-month transaction sequence. Detects: dormant account reactivation for high-value transfers, KYC-to-behaviour mismatch (salary-earner receiving ₹2Cr), sudden volume spikes.
Layer 4 Multi-Signal Risk Fusion + SHAP	Ensemble: graph anomaly score + behavioural deviation score + structuring score + network centrality → single composite risk score (0–100). SHAP generates plain-English causal chain: 'Alert: ₹47.8L split across 6 accounts in 4hrs, reconsolidated at ACCT-5678.'
Layer 5 Investigator Workbench & FIU Report	D3.js animated fund-flow graph, entity relationship map, transaction timeline. Investigator reviews → confirms or dismisses alert → feedback retrains model. Auto-generates STR/CTR evidence package in FIU-IND goAML XML format.

TECHNICAL APPROACH

TECHNOLOGY STACK

Data Ingestion	Apache Kafka, Apache Flink - real-time stream processing
Graph Database	Neo4j / TigerGraph - property graph, Cypher queries
Graph ML	PyTorch Geometric - EvolveGCN, GAT, GraphSAGE
Behavioural ML	TensorFlow LSTM Autoencoder (12-month sequences)
Explainability	SHAP + NLG template engine for narrative alerts
Federated Learning	PySyft / Flower - gradient-only privacy-preserving GNN
Backend API	FastAPI (Python) + Redis cache + PostgreSQL
Frontend	React.js + D3.js - animated transaction graph UI
FIU Integration	goAML API - auto STR/CTR XML package generation
Deployment	Docker + Kubernetes - RBI data localisation compliant

IMPLEMENTATION METHODOLOGY

Phase 1 Data Pipeline	Synthetic fraud dataset generation → Kafka topic setup → graph schema definition → feature engineering. All 5 fraud patterns (layering, round-trip, structuring, dormant, mismatch) seeded.
Phase 2 Model Training	EvolveGCN trained on temporal graphs → LSTM autoencoder on customer sequences → contrastive self-supervised pretraining on unlabelled transactions for label-scarce environments.
Phase 3 Risk Fusion	Ensemble scoring → SHAP causal narrative → threshold calibration per fraud type → alert priority queue → real-time investigator notification with 90-second SLA.
Phase 4 UI & Reporting	D3 animated fund-flow graph → investigator workbench → FIU-IND STR/CTR auto-generation → human-in-the-loop feedback loop that retrains model on confirmed/dismissed alerts.
Phase 5 Federated Layer	PySyft gradient-only sharing across PSB nodes → differential privacy noise injection → RBI data localisation compliance audit → inter-bank fraud signal aggregation (no raw data leaves bank).

FEASIBILITY & VIABILITY

FEASIBILITY ANALYSIS

Technical Readiness

- PyTorch Geometric, Neo4j, Kafka and Flink are all production-grade, open-source and fully documented with active communities
- EvolveGCN and LSTM autoencoders have published benchmarks on the Elliptic and IEEE-CIS financial fraud datasets

Data Availability

- Synthetic fraud dataset generated using configurable templates covering all 5 fraud pattern types for training and testing
- Union Bank CBS, UPI, NEFT/RTGS and card network feeds accessible via existing internal APIs - no new data infrastructure required

Regulatory Alignment

- RBI Master Direction on KYC (2016/2023), PMLA 2002, and FIU-IND goAML format all accommodated in system design
- Federated learning ensures customer transaction data never leaves the originating bank - fully RBI data localisation compliant

CHALLENGES & MITIGATION STRATEGIES

Label scarcity (<0.1% fraud data)

Contrastive self-supervised pretraining on unlabelled transactions learns fraud-relevant embeddings without annotations. Synthetic fraud generation fills remaining gap.

Real-time graph inference at scale

Graph partitioning + Apache Flink stateful streaming + sub-graph inference (not full recompute) keeps alert latency under 800ms even at 10M+ daily transactions.

Adversarial fraud pattern evolution

Built-in red-team simulator generates novel synthetic attack scenarios monthly and stress-tests the model. Auto-retraining triggers when evasion rate exceeds 5%.

Investigator trust in AI-generated alerts

SHAP feature attributions + plain-English causal chain on every alert. Human-in-the-loop feedback loop: confirmed/dismissed alerts retrain the model continuously.

Federated learning coordination cost

Asynchronous gradient averaging with differential privacy noise. IBA already coordinates inter-PSB information sharing - federation builds on existing trust framework.



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AI - CSPARC
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यूनियन बैंक
ऑफ़ इंडिया
भारत सरकार का उपक्रम



Union Bank
of India
A Government of India Undertaking



IMPACT & BENEFITS

₹1.28L Cr

Annual AML losses
avoided (RBI est.)

94%

False positive reduction
vs. rule-based

10 min

FIU report generation
vs. 4 hrs manual

26 PSBs

Addressable banks via
federated network

Social Impact

- Protects citizens whose funds are **laundered** through mule accounts without their knowledge
- Reduces **financial crime** that funds terrorism, drug trafficking, and human trafficking networks
- Builds public trust in PSB digital banking through demonstrably stronger fraud controls
- Enables RBI and FIU-IND to prosecute financial criminals with AI-grade, legally defensible evidence

Economic Impact

- India loses est. ₹1.28 lakh crore annually to financial crime (EY 2023). **GraphSentinel** directly attacks this figure
- Saves ~₹2.4 Cr/year/bank in manual STR/CTR investigator hours through automated report generation
- Fraud-recovered deposits remain in the banking system, improving system-wide credit availability
- Federated network moat: detection accuracy improves with every additional PSB that joins - built-in network effect

Operational Impact

- Investigators handle **3x more cases per day** via AI-prioritised alert queues and auto-drafted evidence packages
- Sub-second alert latency enables fund freezing before layering chain completes across accounts
- Adversarial red-team loop self-hardens the system against emerging fraud patterns without manual rule updates
- Explainable alerts reduce regulator challenge risk - every flag includes SHAP attributions and fund trail narrative

BUSINESS MODEL

VALUE PROPOSITION: Retention-as-Intelligence (RaI)

REVENUE STREAMS

Revenue Stream	Customer	Pricing / Model
SaaS Platform Licence	26 PSBs (IBA members)	₹35 - 75 lakh / bank / year
Alert-Based (RaI) Model	AML Compliance Teams	₹800 per validated catch
Federated Intelligence API	Neo-banks, NBFCs, Fintechs	₹22 / API call + subscription
FIU Integration Module	FIU-IND / Government	₹20-60 lakh institutional licence

TARGET MARKET & SCALABILITY

- Immediate: 12 major PSBs under IBA umbrella - Union Bank, SBI, PNB, Bank of Baroda, Canara Bank, Indian Bank
- Phase 2: 14 remaining scheduled commercial banks + Regional Rural Banks - est. ₹180 Cr ARR at full PSB coverage
- Phase 3: Export to SAARC and ASEAN central banks facing identical AML compliance mandates from FATF
- Global AML software market: USD 4.5 Bn (2024) → USD 9.8 Bn by 2030 (CAGR 13.9%). **GraphSentinel** targets the **explainability + federated intelligence** white-space that **no incumbent** has solved.

GO-TO-MARKET STRATEGY

Month 1-3	Month 4-8	Month 9-18	Month 18+
Deploy pilot at Union Bank fraud investigation unit. 90-day holdout group proves lift vs. existing rule-based system. Baseline metrics established.	IBA partnership onboards 3 additional PSBs into federated network. Cross-bank pattern detection demonstrated with live inter-bank fraud case.	Full 12-PSB rollout + FIU-IND goAML live integration + Global FinTech Fest 2026 investor showcase. Series A fundraise initiated.	International expansion to ASEAN and SAARC banking regulators. Open-source core graph engine released to build global developer community.

RESEARCH & REFERENCES

ACADEMIC REFERENCES

1. Weber et al. (2019) - [Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics](#)
2. Pareja et al. (2020) - [EvolveGCN: Evolving Graph Convolutional Networks for Dynamic Graphs](#)
3. Pourhabibi et al. (2020) - [Fraud Detection: A Systematic Literature Review of Graph-Based Anomaly Detection Approaches](#)
4. Liu et al. (2021) - [Pick and Choose: A GNN-Based Imbalanced Learning Approach for Fraud Detection](#)
5. Feng et al. (2019) - [Hypergraph Neural Networks](#)
6. Chen et al. (2020) - [A Simple Framework for Contrastive Learning of Visual Representations](#)

DATASETS & TOOLS REFERENCED

1. Elliptic Bitcoin Dataset (MIT + Elliptic) - 203,769 txns, 2% fraud; validates GNN approach
2. IEEE-CIS Fraud Detection Dataset (Kaggle 2019) - 590,540 txns; autoencoder baseline
3. FIU-IND goAML Schema Documentation - STR/CTR XML format spec
4. RBI Master Direction on KYC (2016/2023) - customer profiling compliance framework
5. PMLA 2002 & PMLA Rules 2005 - legal basis for ₹10L structuring threshold detection
6. EY India Fraud Survey 2023 - source of ₹1.28L Cr annual loss figure
7. IBA Circular on Sharing of Fraud-Related Information (2023) - federated sharing basis

ONE-PAGE SUMMARY LINK (MANDATORY SUBMISSION)

<https://drive.google.com/file/d/1ADn4EgPfdpY6e6qEouNrEzLrpho6LpSE/view?usp=sharing>